An Introduction to Universal Artificial Intelligence

Chapter 8: Optimality of Universal Agents

Based on Hutter, Quarel & Catt

CRC Press, 2024

*"Perfection is achieved, not when there is nothing more to add,
but when there is nothing left to take away." — Antoine de Saint-Exupéry*

Outline

Why We Need Many Notions of Optimality I

To construct an optimal agent, we first have to say precisely what **optimal** means. This chapter surveys the many definitions of optimality that have been proposed for sequential decision-making agents, compares their relative strengths and weaknesses, and asks whether AIXI (or any other agent) can actually attain them.

Roadmap of the chapter.

- **Section 8.1** defines each optimality criterion in turn, and explains why it is useful and what problems it may have.
- **Section 8.2** shows how a Bayesian agent's behaviour can be made undesirable by a bad choice of prior, and argues why such priors will rarely be used in practice.
- **Section 8.3** explores fundamental obstacles that some of these optimality criteria run into, and under what circumstances optimal policies actually exist.

A quick preview of the five notions we will meet.

- ν (nu)-**optimality**: maximal value in one known, fixed environment ν (nu) – the ideal, but unrealistic for General AI since the true environment is never known in advance.

Why We Need Many Notions of Optimality II

- **Asymptotic optimality**: the agent's value only has to approach the optimal value *eventually*, as time $t \to \infty$. Comes in *strong* (almost sure), *in mean*, *in probability*, and *weak* (Cesàro average) flavours, from strongest to weakest.
- **Bayes-optimality**: maximal expected value under a Bayesian mixture over a whole class of environments – arguably the most reasonable criterion, but not automatically asymptotically optimal.
- **Regret minimization**: comparing lifetime performance to an agent that already knows the true environment; a much stronger, finite-time notion than asymptotic optimality.
- **Pareto optimality**: no other agent does at least as well in every environment and strictly better in at least one. Very weak on its own, but its *balanced* refinement coincides with Bayes-optimality.

8.1.1 Environment-Based (ν-)Optimality I

Environment-based optimality, or ν *(nu)-optimality* for some environment ν, states that a policy is optimal if it achieves the maximum possible value for every history. Recall (Chapter 6) that the value $V^\pi_\nu(h)$ of a policy π in environment ν is the total expected (discounted) future reward, starting from history h.

Definition 8.1.1 (Optimal policy)

A policy π is *optimal* in an environment ν (ν-optimal) iff for all histories, π attains the optimal value:

$$V^\pi_\nu(h) = V^*_\nu(h) \quad \text{for all } h \in (\mathcal{A} \times \mathcal{E})^*.$$

The action a is an *optimal action* iff $a \in \arg\max_{a'} \pi^*_\nu(a'|h)$ for some ν-optimal policy π^*_ν.

*π is the agent's **policy** (its strategy for choosing actions given the history so far); ν is the true **environment** it faces; $V^\pi_\nu(h)$ is the value (expected future reward) that π achieves in ν starting from history h; $V^*_\nu(h)$ is the best value any policy could achieve there; $h = (a_1e_1, \dots)$ is a history of past action/percept pairs; $\mathcal{A}$ and $\mathcal{E}$ are the action and percept spaces. A policy is ν-optimal if it always matches the best possible value.*

8.1.1 Environment-Based (ν-)Optimality II

If we knew the true environment ν in advance, we could choose $\nu = \mu$ and build the agent $\mathrm{AI}\mu$ with policy π^*_ν, which maximizes expected reward over its lifetime – a very natural (if unrealistic) notion of optimality.

Why ν-Optimality Is Too Strong for AGI

The dependence on the **lifetime** or **discount** γ (gamma) could in theory be eliminated by choosing the universal discount $\gamma_t = 2^{-K(t)}$ with infinite horizon (Table 6.4), where $K(t)$ is the Kolmogorov complexity of t (the length of its shortest description). The assumption that the value is a *sum* of *expected* and *non-negative* rewards is fairly mild.

However, the assumption that μ is *known* may only be met in simple artificial settings, such as chess, but is completely unrealistic for AGI purposes – an artificial general intelligence, by definition, does not know its environment in advance. This is exactly why AIXI was introduced: but **AIXI is not** μ-**optimal**. It is only ξ (xi)-**optimal** (optimal with respect to the Bayesian mixture ξ, Section 8.1.3), and it has to *learn* the true environment μ (see Chapter 7).

Conclusion. μ-optimality for the true environment is too narrow / strict a definition of optimality for AGI purposes: no agent can be guaranteed to know the true environment ahead of time. One remedy is to *relax* ν-optimality to the slightly weaker ε (epsilon)-optimality version, defined next.

ε-Optimality and the Heaven-Hell Example

Definition 8.1.2 (ε-optimal policy)

A policy π is ε-*optimal* in an environment ν iff $V_\nu^*(h_{<t}) - V_\nu^\pi(h_{<t}) < \varepsilon$ for all histories $h_{<t} \in \mathcal{H}$.

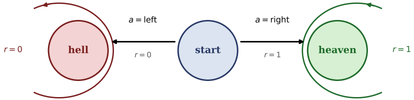
$\varepsilon > 0$ is a small tolerance: the policy π is allowed to fall short of the truly optimal value $V_\nu^*(h_{<t})$ by up to ε, rather than matching it exactly, at every history $h_{<t}$ (all action/percept pairs before time t); $\mathcal{H}$ is the set of all possible histories.

Unfortunately, even this weakened property may be too strong a requirement, since it *still* requires certain prior knowledge of ν. The following example shows that no single policy can be ε-optimal in two sufficiently different environments at once.

Example 8.1.3: The Heaven-Hell Environment I

Let ν_1 be an environment with two actions, *left* and *right*. Taking *left* sends the agent to *hell* forever, receiving the lowest allowed reward $(r = 0)$ at every step; taking *right* sends it to *heaven* forever, receiving the highest allowed reward $(r = 1)$ at every step.

The policy $\pi^*_{\nu_1}$ that always takes $a_1 =$ right is ν_1-optimal. Let ν_2 be identical to ν_1 except that the rewards for left and right are *swapped*. Then $\pi^*_{\nu_1}$ gets the *lowest* reward forever in ν_2 – it is not ν_2-optimal.



ν_1: left → hell forever, right → heaven forever (ν_2 swaps the two)

No policy can be ν_1-optimal *and* ν_2-optimal simultaneously. The best compromise is a random policy $\tilde{\pi}(\epsilon) = \frac{1}{2}$ (choose either action with probability one half), which achieves $V^{\tilde{\pi}}_{\nu_i} = \frac{1}{2} < 1 = V^*_{\nu_i}$ for either i. That is, for $\varepsilon < \frac{1}{2}$, no policy can be ε-optimal in ν_1 and ν_2 simultaneously. ♦

Python: Heaven-Hell Value Computation

Three policies (always-left, always-right, fair coin-flip $\tilde{\pi}(\frac{1}{2})$) evaluated in both ν_1 and ν_2, using per-step reward as the value.

```python
# reward(action, env): nu1: left=0,right=1;  nu2: left=1,right=0
reward = {("left","nu1"):0.0, ("right","nu1"):1.0,
          ("left","nu2"):1.0, ("right","nu2"):0.0}

def value(p, env):    # p = {"left": prob_left, "right": prob_right}
    return sum(pr * reward[(a, env)] for a, pr in p.items())

policies = {"pi*_nu1 (right)": {"left":0.0,"right":1.0},
            "pi*_nu2 (left)":  {"left":1.0,"right":0.0},
            "fair coin":       {"left":0.5,"right":0.5}}

for name, pol in policies.items():
    v1, v2 = value(pol,"nu1"), value(pol,"nu2")
    print(f"{name:16s} V_nu1={v1:.2f} V_nu2={v2:.2f} min={min(v1,v2):.2f}")
```

Output. `pi*_nu1`: $V_{\nu_1} = 1, V_{\nu_2} = 0$; `pi*_nu2`: $V_{\nu_1} = 0, V_{\nu_2} = 1$; `fair coin`: $V_{\nu_1} = V_{\nu_2} = 0.5$ – best worst-case value, so $\varepsilon < \frac{1}{2}$ is unattainable in both simultaneously.

8.1.2 Asymptotic Optimality: The Basic Idea I

At an abstract level, an agent is *asymptotically optimal* if it eventually (asymptotically, i.e. as $t \to \infty$) performs optimally. Mathematically, the value of the agent's policy must converge to the value of the optimal agent, and importantly, this convergence must hold for *every* environment μ in the considered class $\mathcal{M}$.

Definition 8.1.4 (Asymptotic optimality)

A policy π is *asymptotically optimal* in an environment class $\mathcal{M}$ iff for all $\mu \in \mathcal{M}$, the difference between the optimal value function $V^*_\mu(h_{<t})$ and the value function under the policy π, $V^\pi_\mu(h_{<t})$, converges to zero as $t \to \infty$, i.e.,

$$V^*_\mu(h_{<t}) - V^\pi_\mu(h_{<t}) \;\to\; 0 \quad \text{for } t \to \infty \quad \text{a.s.} \mid \text{i.p.} = \text{i.m.} \mid \text{Cesàro}$$

on histories drawn from μ^π.

$\mathcal{M}$ is a whole **class** *of possible environments (not just one); μ^π means the history $h_{<t}$ is generated by running policy π inside environment μ; "a.s." = almost surely, "i.p." = in probability, "i.m." = in mean, and "Cesàro" refers to convergence of the running average – four different senses in which the gap can shrink to zero, discussed below.*

8.1.2 Asymptotic Optimality: The Basic Idea II

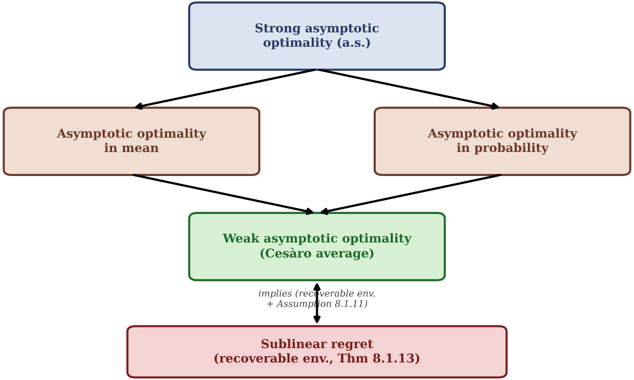
This is the *on-policy* version of **self-optimizingness** (Definition 7.3.3), where the future policy $\tilde{\pi}$ matches the historic policy π – the histories $h_{<t}$ are distributed according to μ^π itself, i.e. the agent evaluates its own trajectory.

Four Flavours of Convergence I

- When convergence is **almost sure** (a.s.), it is called **strong asymptotic optimality**.
- When convergence is **in mean** or **in probability** (i.p.), it is called **asymptotic optimality in mean/probability**.
- When convergence is in **Cesàro average** (the running time-average of the gap goes to zero, even if the gap itself does not), it is called **weak asymptotic optimality**.

Since the logical implications between modes of probabilistic convergence are preserved (Figure 2.8 of the book), **strong** asymptotic optimality implies asymptotic optimality in mean, in probability, and also weak asymptotic optimality. The diagram below summarizes the hierarchy (Section 8.1.4's Theorem 8.1.13 connects the bottom of the hierarchy to regret).

Four Flavours of Convergence II



Weak Asymptotic Optimality

The first (weakest, and hence easiest to satisfy) asymptotic-optimality notion we define carefully is **weak asymptotic optimality**: an agent is weakly asymptotically optimal if its value function converges to the optimal value function *on average*, i.e., convergence in *Cesàro mean*.

Definition 8.1.5 (Weak asymptotic optimality)

A policy π is weakly asymptotically optimal in an environment class $\mathcal{M}$ iff for all $\mu \in \mathcal{M}$,

$$\mathbb{P}^{\pi}_{\mu}\left(\lim_{n\to\infty} \frac{1}{n}\sum_{t=1}^{n}\left[V^{*}_{\mu}(h_{<t}) - V^{\pi}_{\mu}(h_{<t})\right] = 0\right) = 1.$$

$\mathbb{P}^{\pi}_{\mu}(\cdot)$ *is the probability under running policy* π *in environment* μ*; the inner sum* $\frac{1}{n}\sum_{t=1}^{n}[\cdots]$ *is the* **running average** *(over the first* n *time steps) of the instantaneous value gap; requiring this average – rather than each individual gap – to vanish means the agent may make an infinite number of errors, so long as they become rare enough that the average error washes out.*

This means the agent may make infinitely many errors while interacting with the environment, so long as the average number of errors is not too high – the agent will still be weakly asymptotically optimal.

Asymptotic Optimality in Probability / Mean

Convergence in probability and convergence in mean are, in general, *separate* concepts. However, since the difference in value functions, $V^*_\mu(h_{<t}) - V^\pi_\mu(h_{<t})$, is always bounded (by 1, since rewards lie in $[0,1]$) and non-negative (by definition of V^*_μ as the supremum), asymptotic optimality in probability and in mean turn out to be **equivalent** here.

Definition 8.1.6 (Asymptotic optimality in probability/mean)

A policy π is asymptotically optimal in probability (or mean) in an environment class $\mathcal{M}$ iff for all $\mu \in \mathcal{M}$,

$$\lim_{t\to\infty} \mathbf{E}^\pi_\mu \left[V^*_\mu(h_{<t}) - V^\pi_\mu(h_{<t}) \right] = 0.$$

$\mathbf{E}^\pi_\mu[\cdot]$ *is the expectation over histories generated by* π *acting in* μ*; unlike the Cesàro-average criterion above, here the individual (not averaged) expected gap at time* t *must itself vanish as* $t \to \infty$ *– a strictly stronger requirement than weak asymptotic optimality.*

Strong Asymptotic Optimality

Definition 8.1.7 (Strong asymptotic optimality)

A policy π is strongly asymptotically optimal in an environment class $\mathcal{M}$ iff for all $\mu \in \mathcal{M}$,

$$\mathbb{P}_\mu^\pi\Big(\lim_{t\to\infty}\left(V_\mu^*(h_{<t}) - V_\mu^\pi(h_{<t})\right) = 0\Big) = 1.$$

This is the strongest of the four notions: the value gap must converge to zero almost surely along (essentially) every trajectory generated by π in μ – not just on average, and not just in expectation.

As mentioned, strong asymptotic optimality is the *on-policy* version of self-optimizingness. If in Theorem 7.3.6 we choose the historic policy to be the Bayes-optimal policy $\pi = \pi_\xi^*$ itself, then the conclusion of Theorem 7.3.6 is precisely that π_ξ^* is strongly asymptotically optimal – *provided* the theorem's other conditions hold. As we will see in Section 8.3, achieving strong asymptotic optimality in general is exceedingly hard, and in some cases impossible.

8.1.3 Bayesian Optimality

Considered by many to be the most reasonable definition of optimality, **Bayesian optimality** is optimality with respect to a **Bayesian mixture** over a class of environments – rather than requiring optimality in one specific (and possibly unknown) environment.

Definition 8.1.8 (Bayesian optimality)

A policy π is *Bayes-optimal* with respect to prior $\{w_\nu\}$ over environment class $\mathcal{M}$ if for all histories $h_{<t} \in \mathcal{H}$,

$$V^{\pi}_{\xi_w}(h_{<t}) = V^{*}_{\xi_w}(h_{<t})$$

where ξ_w is the Bayesian mixture over $\mathcal{M}$ with prior $\{w_\nu\}$ (Definitions 7.2.1 and 7.2.2).

w_ν is the **prior weight** *the agent assigns to environment ν before seeing any data (all w_ν sum to 1); ξ_w (xi) is the resulting* **Bayesian mixture** *– a single "blended" environment that behaves like a weighted average of every $\nu \in \mathcal{M}$; being Bayes-optimal simply means being ξ_w-optimal in the sense of Definition 8.1.1.*

Key instances. The agent **AIXI** is Bayes-optimal in the class $\mathcal{M}_{sol}$ (all lower semi-computable semimeasures) with the universal prior $w_\nu = 2^{-K(\nu)}$; any agent maximizing **Legg-Hutter intelligence** (Definition 16.7.1) is also Bayes-optimal.

Bayes-Optimality: The Best of Both Worlds

When choosing an optimality criterion, one must be very clear about *what* behaviour is wanted, and *on what timescale*. Asymptotic criteria only guarantee good performance *eventually* – but possibly not before the heat-death of the universe. Regret minimization (Section 8.1.4) works well in the *short term*, but can have undesirable long-run behaviour.

Bayes-optimality can have the best of both worlds: under certain conditions Bayes-optimal agents provably perform well *asymptotically* (Theorems 7.3.1 and 7.3.6), and under the right choice of prior, Bayes-optimal agents can *also* perform well in the short term.

Equivalence to ξ_w-optimality. Bayes-optimality is exactly ξ_w-optimality, a special case of ν-optimality (Definition 8.1.1) applied to the mixture ξ_w itself. The dependency of the definition on the prior w and class $\mathcal{M}$ cannot easily be avoided:

- **Weakening** the requirement (existence of *some* $(w, \mathcal{M})$ making π Bayes-optimal) fails: any policy becomes Bayes-optimal simply by choosing a trivial environment ν_{easy} where every policy gets reward 1 regardless of action, and setting $\mathcal{M} = \{\nu_{easy}\}$.
- **Strengthening** the requirement (Bayes-optimal for *every* choice of $(w, \mathcal{M})$) also fails: given any π, one can construct an *adversarial* environment ν_π that punishes exactly the action π would take. No policy can then be Bayes-optimal for every prior, so the strengthened definition is useless.

Choosing a Good Prior

Bayes-optimality is therefore only *possible* relative to some $(w, \mathcal{M})$, and only *interesting* for "interesting" choices of $\mathcal{M}$ and w – with the pair $\left(2^{-K(\nu)},\ \mathcal{M}_{sol}\right)$ being the most interesting choice from an AGI perspective.

Relying on a prior and model class can be a downside, in the sense that the optimality criterion depends (possibly heavily) on the choice of prior – under different priors, the optimal agent may behave quite differently. Section 8.2 will show that there exist priors that cause a Bayes-optimal agent to act in undesirable ways.

On the other hand, there are good classes and priors for which Bayes-optimal policies *are* self-optimizing (Figure 7.1 in the book). The most promising choice is the **universal prior** $2^{-K_U(\nu)}$, based on a *natural* universal Turing machine U. This pushes the problem to finding a good *natural* Turing machine – a question we return to later in the book.

8.1.4 Regret Minimization

In hindsight it is sometimes easy to look back and see what would have been the best course of action – there are many circumstances where optimal choices only become known *after* the fact. The mathematical form of this notion is called **regret**.

Definition 8.1.9 (Regret)

The *regret* of a policy π in an environment μ with horizon m is

$$\mathrm{Regret}_m(\pi,\mu) := \sup_{\pi'} \mathbf{E}_\mu^{\pi'}\Big[\sum_{t=1}^m r_t\Big] - \mathbf{E}_\mu^{\pi}\Big[\sum_{t=1}^m r_t\Big] = m\Big[\sup_{\pi'} V_{\mu,\gamma=1}^{\pi',m}(\epsilon) - V_{\mu,\gamma=1}^{\pi,m}(\epsilon)\Big]$$

m is the (finite) **horizon** *– how many time steps we look ahead; $\sup_{\pi'} \mathbf{E}_\mu^{\pi'}[\sum_{t=1}^m r_t]$ is the best possible total expected reward over m steps, achieved by some best competing policy π'; the second term is what π itself achieves; regret is simply the* **shortfall** *between the two, rescaled by m to convert the (unnormalized, undiscounted) value into per-step units in the rightmost expression.*

A policy has **sublinear regret** if $\mathrm{Regret}_m(\pi,\mu) = o(m)$, i.e., regret grows strictly slower than the horizon m itself.

Regret vs. Asymptotic Optimality

Since we consider the finite-horizon, unnormalized, undiscounted value ($\gamma_k = [\![k \leq m]\!]$), the regret can be at most $mr_{\max} = m$. While regret is a *performance measure* of the agent, sublinear regret is a *notion of optimality*: an agent has sublinear regret if it does not take suboptimal actions too often.

Although it seems desirable to be as close to optimal as possible, regret minimization has a downside: it makes the agent not care about the long term, since it only cares about value up to horizon m – almost the *exact opposite* problem to asymptotic optimality (which cares only about the long run, and never about the near term).

Example 8.1.10 (Regret is stronger than asymptotic optimality)

Consider an agent π in an environment μ that takes the *least rewarding* action for the first m steps, then performs the μ-optimal action on every step afterward. Measuring performance by regret with horizon m, this agent is *as far from optimal as possible*. However, since it takes μ-optimal actions after a finite time, it is asymptotically optimal. ♦

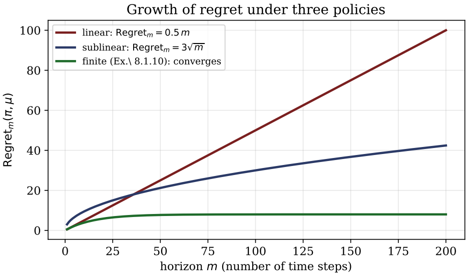
This duality between regret minimization and asymptotic optimality has sparked research into combining the two into a single criterion.

Python: Sublinear vs. Linear vs. Finite Regret

Three qualitatively different regret behaviours: never learning (linear), a reasonable learner (sublinear), and Example 8.1.10's agent (finite regret).

```python
import numpy as np
m = np.arange(1, 201)
linear    = 0.5*m                   # never learns
sublinear = 3.0*np.sqrt(m)          # o(m)
finite    = 8.0*(1-np.exp(-m/15))  # Ex. 8.1.10

for name, r in [("linear", linear),
                ("sublinear", sublinear),
                ("finite", finite)]:
    ratio = r[-1] / m[-1]  # -> 0 iff sublinear
    print(f"{name:9s} R_200={r[-1]:6.2f} ratio={ratio:.3f}")
```



Output. `linear`: ratio$= 0.500$ (does not vanish); `sublinear`: ratio$= 0.212$ (shrinking, i.e. $o(m)$); `finite`: ratio≈ 0.040, shrinking fastest.

Assumptions Needed to Connect Regret and Asymptotic Optimality I

Assumption 8.1.11 (Extra discount function assumptions)

Let the discount function γ (gamma) be such that

- $\gamma_t > 0$ for all t;
- γ_t is decreasing in t;
- $\lim_{t\to\infty} H_t(\varepsilon)/t = 0$ for all $\varepsilon > 0$;

where $H_t(\varepsilon)$ is the *effective horizon* (Definition 6.4.2).

The first bullet says a positive reward is still worth something no matter how far in the future; the second says rewards closer to the present are always worth at least as much as rewards received later; the third bounds how far-sighted the ε-effective horizon $H_t(\varepsilon)$ (roughly: how many future steps still matter) is allowed to be, relative to t itself. **Geometric discounting** *satisfies all three.*

Assumptions Needed to Connect Regret and Asymptotic Optimality II

We also need to restrict to environments the agent can *recover* from mistakes in – a form of *ergodicity* assumption, since an agent exploring an environment it believes is safe may take an action ("fall off a cliff") from which it can never learn to stop repeating.

Let $\mathbf{E}^{\pi_1}_{\nu_1}[V^{\pi_2}_{\nu_2}(h_{<t})]$ denote the expected value function of policy π_2 on environment ν_2, given a past history $h_{<t}$ generated by policy π_1 on environment ν_1.

Recoverable Environments

Definition 8.1.12 (Recoverable environments)

An environment μ is *recoverable* iff

$$\sup_{\pi} \left| \mathbf{E}_{\mu}^{\pi_{\mu}^{*}} \left[V_{\mu}^{*}(h_{<t}) \right] - \mathbf{E}_{\mu}^{\pi} \left[V_{\mu}^{*}(h'_{<t}) \right] \right| \longrightarrow 0 \quad \text{for } t \to \infty$$

Note the expectations are with respect to different policies: the history $h_{<t}$ is sampled from running the optimal policy π_{μ}^{*}, whereas $h'_{<t}$ is sampled from running an arbitrary policy π. Intuitively, **recoverable** *means that no matter what the past history was, or how poor a policy π was followed, switching to an optimal policy π^{*} from time t onward converges to the same value as if the optimal policy had been followed from the very beginning.*

This means that no action can permanently lock the agent out of parts of the environment needed for maximum reward, and any early mistakes can be recovered from – this excludes things like cliffs one cannot climb back from, and traps such as the Heaven-Hell environment (Example 8.1.3), where taking *left* once locks the agent into hell forever.

Theorem 8.1.13: Sublinear Regret in Recoverable Environments

Theorem 8.1.13 (Sublinear regret in recoverable environments)

If the discount function γ satisfies Assumption 8.1.11, the environment μ is recoverable, and π is asymptotically optimal in mean in $\mathcal{M} = \{\mu\}$, then $\mathrm{Regret}_m(\pi, \mu) = o(m)$.

In words: if the discount is well-behaved, the environment lets you recover from mistakes, and your policy is asymptotically optimal in mean (Definition 8.1.6), then your policy is automatically guaranteed to have sublinear regret. This is one concrete way of combining the "short-term" regret criterion with the "long-term" asymptotic-optimality criterion.

This connects the middle of the hierarchy figure shown earlier (asymptotic optimality in mean) to sublinear regret at the bottom, under the recoverability and discount assumptions. The full proof (next slides) is a careful bookkeeping argument combining the triangle inequality with properties of the effective horizon.

Proof of Theorem 8.1.13 I

We follow the proof from [Lei16b]. Let $\pi_m := \arg\max_\pi \mathrm{Regret}_m(\pi, \mu)$. We want to show that

$$\limsup_{m\to\infty} \mathbf{E}_\mu^{\pi_m}\Big[\frac{1}{m}\sum_{t=1}^m r_t\Big] - \mathbf{E}_\mu^{\pi}\Big[\frac{1}{m}\sum_{t=1}^m r_t\Big] \leq 0.$$

Let $d_k^{(m)} := \mathbf{E}_\mu^{\pi_m}[r_k] - \mathbf{E}_\mu^{\pi}[r_k]$. We have $-1 \leq d_k^{(m)} \leq 1$, since rewards are bounded between 0 and 1. From the definition of *recoverable*, since π and π_m are not worse than the worst policy, we have that

$$\Big|\mathbf{E}_\mu^{\pi_\mu^*}[V_\mu^*(h_{<t})] - \mathbf{E}_\mu^{\pi}[V_\mu^*(h_{<t})]\Big| \to 0 \quad \text{and} \quad \sup_m \Big|\mathbf{E}_\mu^{\pi_\mu^*}[V_\mu^*(h_{<t})] - \mathbf{E}_\mu^{\pi_m}[V_\mu^*(h_{<t})]\Big| \to 0$$

for $t \to \infty$. Combining these with the triangle inequality we get

$$\sup_m \mathbf{E}_\mu^{\pi_m}[V_\mu^*(h_{<t})] - \mathbf{E}_\mu^{\pi}[V_\mu^*(h_{<t})] \to 0 \quad \text{for } t \to \infty. \tag{8.1.14}$$

Since π is asymptotically optimal in mean we have that

$$\mathbf{E}_\mu^{\pi}[V_\mu^*(h_{<t})] - \mathbf{E}_\mu^{\pi}[V_\mu^{\pi}(h_{<t})] \to 0 \quad \text{for } t \to \infty.$$

Proof of Theorem 8.1.13 II

Combining the two previous lines we get

$$\sup_m \mathbf{E}_\mu^{\pi_m}[V_\mu^*(h_{<t})] - \mathbf{E}_\mu^{\pi}[V_\mu^{\pi}(h_{<t})] \to 0 \quad \text{for } t \to \infty.$$

Since $V_\mu^* \geq V_\mu^{\pi_m}$ we have

$$\limsup_{t\to\infty} \left[\sup_m \mathbf{E}_\mu^{\pi_m}[V_\mu^{\pi_m}(h_{<t})] - \mathbf{E}_\mu^{\pi}[V_\mu^{\pi}(h_{<t})] \right] \leq 0. \tag{8.1.15}$$

Proof of Theorem 8.1.13 III

For any policy π' we can rewrite our expectation of the value function as

$$E_\mu^{\pi'}[V_\mu^{\pi'}(h_{<t})] = \mathbf{E}_\mu^{\pi'}\left[\frac{1}{\Gamma_t}\sum_{k=t}^{\infty}\gamma_k r_k\right] = \frac{1}{\Gamma_t}\sum_{k=t}^{\infty}\gamma_k \mathbf{E}_\mu^{\pi'}[r_k]$$

Then, using this form for (8.1.14) and (8.1.15) and $d_k^{(m)}$, we get

$$\limsup_{t\to\infty}\sup_m \frac{1}{\Gamma_t}\sum_{k=t}^{\infty}\gamma_k d_k^{(m)} \leq 0.$$

Let $\varepsilon > 0$. Choose t_0 independent of m and large enough such that $\frac{1}{\Gamma_t}\sum_{k=t}^{\infty}\gamma_k d_k^{(m)} < \varepsilon$ for all m and $t \geq t_0$. We split $\sum_{k=1}^{m} d_k^{(m)}$ at t_0, and define b_t for $t_0 \leq t \leq m$ as

$$b_{t_0} = \frac{\Gamma_{t_0}}{\gamma_{t_0}} \qquad \text{and} \qquad b_t = \frac{\Gamma_t}{\gamma_t} - \frac{\Gamma_t}{\gamma_{t-1}} \quad \text{for } t > t_0.$$

Proof of Theorem 8.1.13 IV

The following equalities for b_t can be verified by inserting the definitions and swapping the two sums:

$$\sum_{t=t_0}^{m} d_t^{(m)} = \sum_{t=t_0}^{m} \frac{b_t}{\Gamma_t} \sum_{k=t}^{m} \gamma_k d_k^{(m)} \qquad \text{and} \qquad \sum_{t=t_0}^{m} \frac{b_t}{\Gamma_t} = \frac{1}{\gamma_m}. \tag{8.1.16}$$

Proof of Theorem 8.1.13 V

Additionally, for the sum from t_0 to m of b_t we get

$$\sum_{t=t_0}^{m} b_t = \frac{\Gamma_{t_0}}{\gamma_{t_0}} + \sum_{t=t_0+1}^{m} \left(\frac{\Gamma_t}{\gamma_t} - \frac{\Gamma_t}{\gamma_{t-1}}\right) = \frac{\Gamma_{m+1}}{\gamma_m} + m - t_0 + 1$$

From the definition of the effective horizon $H_m(\varepsilon)$ (Definition 6.4.2) and the monotonicity assumption on γ we get

$$\varepsilon\Gamma_m \geq \Gamma_{m+H_m(\varepsilon)} = \Gamma_m - \gamma_m - \cdots - \gamma_{m+H_m(\varepsilon)-1} \geq \Gamma_m - H_m(\varepsilon)\gamma_m$$

which implies $\quad \dfrac{H_m(\varepsilon)}{1-\varepsilon} \geq \dfrac{\Gamma_m}{\gamma_m} \geq \dfrac{\Gamma_{m+1}}{\gamma_m}.$

Proof of Theorem 8.1.13 VI

Combining everything we get for the regret

$$\begin{aligned}
\mathrm{Regret}_m(\pi,\mu) = \sum_{t=1}^{m} d_t^{(m)} &\le t_0 + \sum_{t=t_0}^{m} b_t \frac{1}{\Gamma_t} \sum_{k=t}^{m} \gamma_k d_k^{(m)} \\
&\le t_0 + \sum_{t=t_0}^{m} b_t \frac{1}{\Gamma_t} \sum_{k=t}^{\infty} \gamma_k d_k^{(m)} - \sum_{t=t_0}^{m} \frac{b_t}{\Gamma_t} \sum_{k=m+1}^{\infty} \gamma_k d_k^{(m)} \\
&< t_0 + \sum_{t=t_0}^{m} b_t \varepsilon + \sum_{t=t_0}^{m} \frac{b_t \Gamma_{m+1}}{\Gamma_t} \\
&= t_0 + \varepsilon \frac{\Gamma_{m+1}}{\gamma_m} + \varepsilon(m - t_0 + 1) + \frac{\Gamma_{m+1}}{\gamma_m} \\
&\le t_0 + \frac{(1+\varepsilon) H_m(\varepsilon)}{1-\varepsilon} + \varepsilon(m - t_0 + 1)
\end{aligned}$$

By Assumption 8.1.11 we have $H_m(\varepsilon) = o(m)$, hence

$$\limsup_{m\to\infty} \frac{1}{m} \mathrm{Regret}_m(\pi,\mu) \le \varepsilon.$$

Since ε was arbitrary, we have the desired result. ■

8.1.5 Pareto Optimality I

Related to the game-theoretic notion of **Pareto Efficiency**, an agent is *Pareto optimal* if there is no other agent which is at least as good in *every* environment, and strictly better in *at least one*.

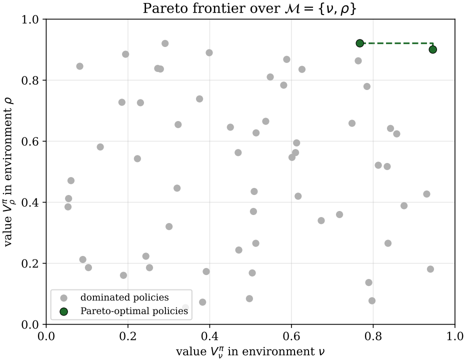
Definition 8.1.17 (Pareto optimality)

A policy π is Pareto optimal in a set of environments $\mathcal{M}$ iff there is no policy π' such that $V_\nu^{\pi'} \geq V_\nu^\pi$ for all $\nu \in \mathcal{M}$ and $V_\rho^{\pi'} > V_\rho^\pi$ for at least one $\rho \in \mathcal{M}$.

π' "dominates" π if it never does worse in any environment $\nu \in \mathcal{M}$, and does strictly better in at least one environment $\rho \in \mathcal{M}$; π is Pareto optimal exactly when no such dominating π' exists.

Pareto optimality is useful when there is a *set of conflicting criteria* we want to be optimal in simultaneously – here, the different criteria are performance in the different environments $\nu \in \mathcal{M}$. The set of Pareto-optimal agents is called the **Pareto frontier**.

8.1.5 Pareto Optimality II



Theorem 8.1.18: AIξ and AIXI Are Pareto Optimal I

Theorem 8.1.18 (AIξ and AIXI are Pareto optimal)

The Bayes-optimal policy π^*_ξ (and in particular AIXI, its special case with $\mathcal{M} = \mathcal{M}_{sol}$) is Pareto optimal.

Proof. Assume π^*_ξ is *not* Pareto optimal, and let π be a dominating policy, i.e. $V^\pi_\nu \geq V^{\pi^*_\xi}_\nu$ for all $\nu \in \mathcal{M}$ with strict inequality for at least one $\nu \in \mathcal{M}$. Then

$$V^\pi_\xi = \sum_\nu w_\nu V^\pi_\nu > \sum_\nu w_\nu V^{\pi^*_\xi}_\nu = V^{\pi^*_\xi}_\xi \equiv V^*_\xi \geq V^\pi_\xi$$

The two equalities follow from **linearity** *of V_ρ (Theorem 7.2.5, the mixture value is a weighted sum of the individual values); the strict inequality follows from the domination assumption and $w_\nu > 0$ for all ν; the last inequality follows because π^*_ξ maximizes value under the mixture ξ by definition. The chain gives $V^\pi_\xi > V^\pi_\xi$ – a contradiction – proving Pareto optimality of $\mathrm{AI}\xi$. AIXI is the special case of $\mathrm{AI}\xi$ with $\mathcal{M} = \mathcal{M}_{sol}$.*

■

Theorem 8.1.18: AIξ and AIXI Are Pareto Optimal II

It is comforting that $\mathrm{AI}\xi$ is Pareto optimal, since a violation would clearly render π^*_ξ intuitively suboptimal. Unfortunately, for any class of environments containing $\mathcal{M}_{comp}$ (all computable environments), *every* policy is Pareto optimal [Lei16b] – so Pareto optimality becomes a **vacuous** notion for AIXI.

Theorem 8.1.19: Pareto Optimality Becomes Vacuous I

Theorem 8.1.19 (Every policy is Pareto optimal in any $\mathcal{M} \supseteq \mathcal{M}_{comp}$)

If $\mathcal{M} \supseteq \mathcal{M}_{comp}$ (contains all computable environments), then every policy π is Pareto optimal in $\mathcal{M}$.

Proof (sketch). [Lei16b] Assume for contradiction that some policy π' strictly dominates π: $V_\nu^{\pi'} \geq V_\nu^\pi$ for all ν, with strict inequality at some $\rho \in \mathcal{M}$. Since $V_\rho^{\pi'} > V_\rho^\pi$, there must be some history where π and π' take different actions. Let $\dot{h}_{<t}$ be the (lexicographically) *first* such history, and $\dot{a}_t$ an action with $\pi(\dot{a}_t|\dot{h}_{<t}) > \pi'(\dot{a}_t|\dot{h}_{<t})$.

We can define a *computable* environment μ, identical to ρ for any history that is not an extension of $\dot{h}_{<t}$, but which rewards 1 forever if action $\dot{a}_t$ is taken at $\dot{h}_{<t}$, and 0 otherwise. Since π takes $\dot{a}_t$ more likely than π' given $\dot{h}_{<t}$, and they coincide otherwise, we get $V_\mu^\pi > V_\mu^{\pi'}$ – contradicting that π' dominates π, since $\mu \in \mathcal{M}_{comp} \subseteq \mathcal{M}$. ■

Takeaway. In the context of Universal AI, Pareto optimality gives no useful information: virtually *any* policy, however bad, satisfies it – so we need a *refined* version.

Balanced Pareto Optimality I

Definition 8.1.20 (Balanced Pareto optimality [Hut02c, Lei16b])

Given a prior $w \in \Delta'\mathcal{M}$ where $\mathcal{M}$ is an environment class, a policy π is *balanced Pareto optimal* if it achieves a better value across $\mathcal{M}$ *weighted by* $w \in \Delta'\mathcal{M}$ than any other policy, formally, if for all policies $\tilde{\pi}$,

$$\sum_{\nu \in \mathcal{M}} w_\nu \left[V_\nu^\pi - V_\nu^{\tilde{\pi}} \right] \geq 0.$$

*Instead of demanding domination in every single environment (too weak, as we saw), balanced Pareto optimality only requires domination in the prior-***weighted average** *sense – exactly analogous to how Bayesian optimality relaxes ν-optimality.*

All and only Bayes-optimal policies $\pi^*_{\xi_w}$ are balanced Pareto optimal for w:

$$\sum_{\nu \in \mathcal{M}} w_\nu [V_\nu^\pi - V_\nu^{\tilde{\pi}}] = V_{\xi_w}^\pi - V_{\xi_w}^{\tilde{\pi}} \geq 0 \;\; \forall \tilde{\pi} \quad \text{iff} \quad \pi = \pi^*_{\xi_w}$$

Balanced Pareto Optimality II

Hence balanced Pareto optimality has the *same* features (advantages and problems) as Bayes-optimality (Section 8.1.3), e.g. its dependence on the prior. Section 8.2 describes possible bad priors that make balanced Pareto optimality a less useful definition.

8.2 Bad Priors: Overview

The prior is one of the most important aspects for Bayesian optimality. However, as we will show below, some priors are less than ideal, and can cause Bayes-optimal policies to perform quite poorly [Ors10, Lei16b, LH15b].

We consider two families of “bad” priors:

- the **dogmatic prior** (Section 8.2.1): makes the Bayes-optimal agent dogmatically follow a single, arbitrary, pre-chosen policy;
- the **indifference prior** (Section 8.2.2): makes *every* policy Bayes-optimal, so the agent’s actions are determined purely by an arbitrary tie-breaking rule.

Both priors are *adversarially chosen* – they are constructed specifically to cause bad behaviour, and are highly unnatural.

8.2.1 The Dogmatic Prior

Given any policy and a history generated by that policy interacting with the environment, there is a prior for which *following that policy* is Bayes-optimal. We call this the **dogmatic prior**, as the Bayes-optimal agent *dogmatically* believes it must follow that policy. Dogmatically following a single policy will rarely be what we want a Bayes-optimal agent to do, since the agent will never learn from its mistakes – but note the priors here are adversarially chosen.

Theorem 8.2.1 (Dogmatic prior [LH15b])

Let π be any deterministic computable policy, let ξ be any Bayesian mixture over $\mathcal{M}_{sol}$ with weights $w \in \Delta'\mathcal{M}_{sol}$, and let $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$. There is a universal mixture ξ' with respect to some prior such that for any history $h_{<t}$ consistent with π and $V_\xi^\pi(h_{<t}) > \varepsilon$, the action $\pi(h_{<t})$ is the *unique* ξ'-optimal action.

$\Delta'\mathcal{M}_{sol}$ is the set of valid priors (probability distributions) over $\mathcal{M}_{sol}$; the theorem says we can build a new universal mixture ξ' under which any fixed deterministic policy π becomes the uniquely optimal thing to do – for as long as π's own value estimate stays above the threshold ε.

Proof of Theorem 8.2.1 I

This proof follows [Lei16b]. For all environments $\nu \in \mathcal{M}_{sol}$ we can define a *new* environment $\rho_{\pi,\nu} \in \mathcal{M}_{sol}$ that is identical to ν until it receives an action π would *not* take, then outputs reward 0 forever. We derive new weights w' that weigh environments $\rho_{\pi,\nu}$ higher than ν. Define

$$w'_\nu := \varepsilon w_\nu \text{ if } \nu \neq \rho_{\pi,\nu'} \,\forall \nu' \in \mathcal{M}_{sol} \qquad \text{and} \qquad w'_{\rho_{\pi,\nu}} := (1-\varepsilon) w_\nu + \varepsilon w_{\rho_{\pi,\nu}} \text{ otherwise}$$

Checking that the new weights sum to 1,

$$\begin{aligned}
\sum_{\nu \in \mathcal{M}_{sol}} w'_\nu &= \sum_{\nu = \rho_{\pi,\nu}} w'_\nu + \sum_{\nu \neq \rho_{\pi,\nu'} \forall \nu'} w'_\nu \\
&= \sum_{\nu = \rho_{\pi,\nu}} \big((1-\varepsilon) w_{\rho_{\pi,\nu}} + \varepsilon w_\nu\big) + \sum_{\nu \neq \rho_{\pi,\nu'} \forall \nu'} \varepsilon w_\nu \\
&= \sum_{\nu \in \mathcal{M}_{sol}} \varepsilon w_\nu + \sum_{\nu \in \mathcal{M}_{sol}} (1-\varepsilon) w_\nu = \varepsilon + (1-\varepsilon) = 1
\end{aligned}$$

Therefore w' is a valid prior over $\mathcal{M}_{sol}$, and we define a Bayesian mixture ξ' using w'. We can also define the mixture over the ρ's by $\rho := \sum_{\nu \in \mathcal{M}_{sol}} w_\nu \rho_{\pi,\nu}$, and rewrite ξ' as $\xi' = \varepsilon \xi + (1-\varepsilon)\rho$.

Proof of Theorem 8.2.1 II

From now on, let $h_{<t}$ be a history consistent with π, i.e. actions generated by π. For such $h_{<t}$, $\rho_{\pi,\nu} = \nu$, hence $\rho = \xi$, hence $\xi' = \xi$. Considering the class $\mathcal{M} = \{\xi, \rho\}$ with prior $w_\xi = \varepsilon$ and $w_\rho = 1 - \varepsilon$, this implies that the posterior coincides with the prior on such $h_{<t}$ (Definition 7.2.2). Linearity of the Q-value function (Theorem 7.2.5) for any policy π' and action a_t can hence be expressed in terms of prior weights

$$Q_{\xi'}^{\pi'}(h_{<t}a_t) = \varepsilon Q_\xi^{\pi'}(h_{<t}a_t) + (1-\varepsilon) Q_\rho^{\pi'}(h_{<t}a_t)$$

Let $a^* = \pi(h_{<t})$ and a' be any other action. We show that a^* is the ξ'-optimal action by showing $V_{\xi'}^*(h_{<t}a^*) > V_{\xi'}^*(h_{<t}a')$.

$$Q_{\xi'}^*(h_{<t}a^*) \geq Q_{\xi'}^\pi(h_{<t}a^*) = Q_\xi^\pi(h_{<t}a^*) = V_\xi^\pi(h_{<t}) > \varepsilon$$

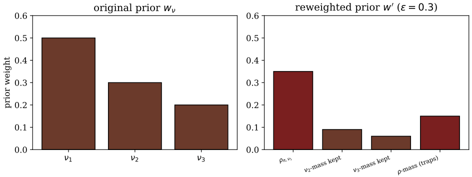
$$Q_{\xi'}^*(h_{<t}a') = \varepsilon Q_\xi^\pi(h_{<t}a') + (1-\varepsilon) Q_\rho^\pi(h_{<t}a') = \varepsilon Q_\xi^\pi(h_{<t}a') + 0 \leq \varepsilon$$

Therefore $a^* = \pi(h_{<t})$ is the ξ'-optimal action for history $h_{<t}$. ■

Python: Dogmatic-Prior Weight Redistribution

Toy illustration of Theorem 8.2.1's weight redistribution: a 3-environment class $w = (0.5, 0.3, 0.2)$, $\varepsilon = 0.3$; following π (consistent with ν_1) traps ν_2, ν_3-mass into ρ.

```python
w = {"nu1":0.5, "nu2":0.3, "nu3":0.2}
eps = 0.3
# nu1 consistent with pi -> rho=nu1 here
w_prime = {
  "rho_pi_nu1": (1-eps)*w["nu1"]+eps*w["nu1"],
  "nu2_kept":   eps*w["nu2"],
  "nu3_kept":   eps*w["nu3"],
}
trapped = (1-eps)*(w["nu2"]+w["nu3"])
print("w'       :", w_prime)
print("trapped :", round(trapped,3))
print("total    :", round(sum(w_prime.values())+
      trapped,3))
```



Output. w' still sums to 1.0, but now favours the π-consistent branch ($\rho_{\pi,\nu_1} = 0.5$), while deviating from π routes mass into trap environments rewarding 0 forever.

8.2.2 The Indifference Prior

Another bad prior is the **indifference prior**. Under certain conditions, this prior makes *all* policies Bayes-optimal with respect to it. Because all policies are optimal (and therefore equal in value), the Bayes-optimal agent using this prior chooses its actions entirely based on whatever **tie-breaker** is used – which is not ideal, since depending on the tie-breaker, the agent can be convinced to take *any* arbitrary action.

Theorem 8.2.2 (Indifference prior [LH15b])

If there is an m such that the discount normalization factor $\Gamma_{m+1} = 0$, then there is a Bayesian mixture ξ' such that **all** policies are ξ'-optimal.

Γ_{m+1} *(Gamma) is the discount normalizer for time steps after* $m+1$ *(Table 6.4); if it is exactly zero, it means nothing that happens after time* m *can affect the value function at all, so the mixture* ξ' *can be constructed to render every policy's choices after* m *irrelevant, while "masking" its earlier choices too.*

This relies heavily on the assumption that eventually Γ_{m+1} becomes 0, which is never the case with the standard **geometric discounting** (where $\Gamma_t > 0$ for all finite t) – but does happen for finite-lifetime discounting.

Proof of Theorem 8.2.2 I

This proof follows [Lei16b]. Assume the action space is $\mathcal{A} = \{0, 1\}$. Let U be some universal Turing machine and define U' such that U' does not halt on programs of length less than m, and for programs of length greater than $m-1$,

$$U'(p, a) := U(p_{m:\ell(p)}, \; a \text{ xor } p_{1:\ell(a)})$$

$\ell(p)$ is the length of program p; "xor" flips bits of the action according to the first $\ell(a)$ bits of the program p – so U' reserves its first $m-1$ bits purely to mask (via XOR) the actions taken, rather than using them for anything else.

Then we can define a chronological Solomonoff distribution (which by Theorem 3.8.8, extended to the agent case (7.4.9), is a Bayesian mixture ξ') with respect to U' as the underlying universal Turing machine as

$$\xi'(e_{<m} \| a_{<m}) = \sum_{p : e_{<m} \sqsubseteq U'(p, a_{<m})} 2^{-\ell(p)} \stackrel{(a)}{=} \sum_{s_{<m} p' : e_{1:t} \sqsubseteq U'(s_{<m} p', a_{<m})} 2^{-m-1-\ell(p')}$$

$$= \sum_{s_{<m}} \sum_{p' : e_{1:t} \sqsubseteq U(p', a_{<m} \text{ xor } s_{<m})} 2^{-m-1-\ell(p')} \stackrel{(b)}{=} \sum_{s_{<m}} \sum_{p' : e_{1:t} \sqsubseteq U(p', s_{<m})} 2^{-m-1-\ell(p')}$$

Proof of Theorem 8.2.2 II

Step (a) comes from decomposing p as $s_{<m}p$. Step (b) comes from the fact that we are summing over *all* $s_{<m}$ and xor-ing them with $a_{<m}$, which is identical to simply summing over all $s_{<m}$ again (relabelling the dummy variable). Therefore the mixture ξ' does **not** depend on $a_{<m}$ for the first $m-1$ actions.

For the remaining actions, since $\Gamma_m = 0$, the rewards after time step m do not matter. Since the mixture does not depend on the actions taken (for the first $m-1$ steps) and rewards after m are irrelevant, *all* policies have the same value V_ξ^π – hence all policies are optimal with respect to this mixture. ■

Remark. This example is very artificial, and not unexpected given the unnatural universal Turing machine U', which depends on m and reserves $m-1$ bits to "mask" the first $m-1$ actions. If we increase AIXI's lifetime while fixing the UTM U', the result no longer holds.

Discussion: Solomonoff Induction Errors

An analogous result holds for **Solomonoff induction**: Theorem 3.8.9 and Corollary 3.5.14 imply that Solomonoff's distribution M makes at most

$$K(\mu) \cdot 2\ln 2 = Km(x_{1:\infty}) \cdot 2\ln 2$$

errors for predicting deterministic sequences $x_{1:\infty}$.

$K(\mu)$ is the Kolmogorov complexity of the true generating environment μ; $Km(x_{1:\infty})$ is (essentially) the same quantity phrased in terms of the sequence's monotone complexity; $2\ln 2$ is a fixed constant coming from the error-bound proof. In words: Solomonoff induction makes a bounded number of prediction errors, proportional to how "complex" (hard to compress) the true sequence is.

In case the shortest program (for μ) has length $> m$, there is **no guarantee** that we make fewer than m errors – so the indifference-prior pathology is really about the interaction between a *finite* lifetime m and an adversarially long minimal program length, not a failure of Solomonoff induction itself.

8.2.3 Bad Priors, Bad Agents

Clearly, when the above “bad priors” are used, the Bayes-optimal agent is *still* Bayes-optimal – but in both cases it does not behave as we would like. Intuitively, we know that following an arbitrary fixed policy, or being indifferent to all actions, are both *not* generally intelligent behaviours.

Does this mean Bayesian optimality is a poor choice of optimality criterion?

No. The existence of a bad prior does *not* invalidate the idea of Bayesian optimality – it just means one has to be **careful** when choosing a prior. Importantly, the existence of bad priors does *not* mean good priors do not exist, or are not easy to find. Indeed, it is plausible that *natural* UTMs lead to good priors (see also Example 2.7.5).

This is why the choice of a good, *natural* universal Turing machine matters so much for the universal prior $2^{-K_U(\nu)}$ discussed in Section 8.1.3 – a theme the book returns to.

8.3 Problems with Optimality Criteria

The quest for a "perfect" optimality criterion – one strong enough to be *useful*, and weak enough that it can actually be *satisfied* – may in fact be futile. In this section we explore the weaknesses and feasibility of the optimality notions introduced in Section 8.1.

We want an optimality criterion with several properties simultaneously:

- eventually performing well and **not falling into traps** (generally, regret-minimizing agents tend to *avoid* traps, but asymptotically optimal agents may *jump into* traps in the name of exploration);
- agents satisfying the criterion should be somewhat **practical**, since we would like to eventually implement such agents in the real world.

This is exactly where criteria such as asymptotic optimality fall short: for an arbitrarily long amount of time, an asymptotically optimal agent may receive low or no reward [Lei16b].

Theorem 8.3.1: No Deterministic Strong Asymptotically Optimal Policy I

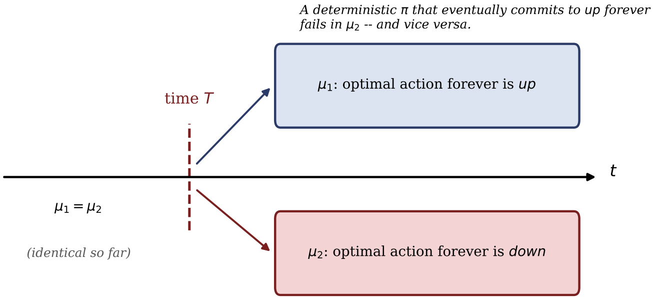
Theorem 8.3.1 (No deterministic strong asymptotically optimal policy [LH11a])

There is no deterministic policy that is strongly asymptotically optimal (Definition 8.1.7) in the class of environments $\mathcal{M} \supseteq \mathcal{M}_{comp}$.

Proof sketch. By contradiction. Assume there is a deterministic strongly asymptotically optimal policy π. Construct two computable environments $\mu_1, \mu_2 \in \mathcal{M}$ (since $\mathcal{M} \supseteq \mathcal{M}_{comp}$) that are identical up to a time T, at which point the unique optimal action in μ_1 is *up* forever, and the unique optimal action in μ_2 is *down* forever.

Since π is strongly asymptotically optimal in μ_1, there exists T such that for all $t \geq T$, $\pi(h_{<t}) = up$ forever – but this means π is *not* strongly asymptotically optimal in μ_2, a contradiction. Hence no deterministic strongly asymptotically optimal policy exists. ■

Theorem 8.3.1: No Deterministic Strong Asymptotically Optimal Policy II



Theorem 8.3.2: No Weak Asymptotically Optimal Policy for General $\mathcal{M}$

Theorem 8.3.1 is a very general result – since it includes all deterministic policies, it also rules out the Bayes-optimal policy AIXI. We can be even more general and show that there does not exist *any* policy (not even a randomized one) achieving even the *weakest* asymptotic-optimality notion, for the class of *all* environments.

Theorem 8.3.2 (No weak asymptotically optimal policy for general $\mathcal{M}$)

If we consider $\mathcal{M}$ to be the class of *all* environments, then there is no weakly asymptotically optimal policy for $\mathcal{M}$.

Proof. This comes from constructing an environment specifically *tuned against* a given policy π – one where the reward for actions is chosen such that there will be the *largest* difference between the value of actions taken by π and the optimal actions. Since there are no restrictions on the computability level of the environments (unlike Theorem 8.3.1's $\mathcal{M}_{comp}$), this construction is possible for *any* policy π, computable or not. ■

Python: The Adversarial Time-T Switch

A minimal simulation of the Theorem 8.3.1 construction: two environments agree up to time T, then require opposite actions forever.

```python
# correct action after T: "up" for mu1, "down" for mu2
correct = {"mu1": "up", "mu2": "down"}

# a deterministic policy eventually committing to a fixed action
committed_action = "up"

for mu, want in correct.items():
    v = 1.0 if committed_action == want else 0.0
    print(f"committed to '{committed_action}' -> value in {mu} = {v}")
```

Output. value in `mu1` $= 1.0$ (matches); value in `mu2` $= 0.0$ (forever wrong). A deterministic policy that is strongly asymptotically optimal in μ_1 must eventually commit to one fixed action, so it necessarily fails in μ_2 – Theorem 8.3.1's contradiction.

Where Does This Leave Us?

The impossibility results above paint a bleak picture for the notion of asymptotic optimality for general history-based agents. But the situation is not as bad as it looks:

- The proofs crucially depend on the policy being **deterministic**. **Weak** (Theorem 9.4.1) and **strong** (Theorem 9.4.2) asymptotic optimality are actually *achievable* by **stochastic** policies that add extra exploration to $\mathrm{AI}\xi$ (Section 9.4).

Open questions. Which optimality criterion an artificial general intelligence would or should satisfy remains an open question:

- Should we conclude from the fact that $\mathrm{AI}\xi$ is *not* strongly asymptotically optimal that we should use Inq (an exploratory variant), which *is*?
- Or does Inq fare worse than $\mathrm{AI}\xi$ with respect to *other*, possibly more important, optimality criteria?
- For instance, the **price** of asymptotic optimality is *certain death* [CHC21] – achieving it may require taking arbitrarily risky exploratory actions.

Studying how these criteria relate to each other, and which are achievable under which conditions, helps narrow down desirable optimality criteria for both theoretical and practical artificial general intelligences.

8.4 Exercises I

1. [C10] Prove that an agent is balanced Pareto optimal iff it is Bayes-optimal.
2. [C20] Prove that strong asymptotic optimality implies both asymptotic optimality in mean and also weak asymptotic optimality.
3. [C20] Give an example of an environment class $\mathcal{M}$ and a family of policies $\{\pi_i\}$ such that no policy is Pareto optimal.
4. [C20] Prove that geometric discounting satisfies Assumption 8.1.11, and that all of the other discounting methods (power, finite lifetime, harmonic, hyperbolic, universal, no discount) from Table 6.4 fail at least one of the three assumptions.
5. [C15] Prove the equalities in (8.1.16).
6. [C24] Prove that if $\mathcal{A}$, $\mathcal{E}$ are finite, a Pareto optimal policy exists.
7. [C19] Prove that if π is not Pareto optimal, then it is Pareto dominated by a Pareto optimal policy π'.
8. [C24] Weaken the premises of Theorem 8.2.1 to no longer require that $V_\xi^\pi(h_{<t}) > \varepsilon$. Prove that the action $\pi(h_{<t})$ is a ξ'-optimal action, but may not be uniquely optimal.

8.4 Exercises II

9. [C15] Show that for all π, there exists an environment ν such that π is not ν-optimal.
10. [C30] Formalize the proof of Theorem 8.3.1.
11. [C18] Formalize the proof of Theorem 8.3.2.
12. [C28] Provide a counter-example that convergence-in-mean does not imply Cesàro-convergence almost surely.
13. [C20] Prove that convergence-in-mean implies Cesàro convergence-in-mean.
14. [C23] Prove convergence almost surely implies Cesàro convergence almost surely.
15. [C18] Provide a counter-example that weak asymptotic optimality does not imply asymptotic optimality in mean and a counter example that asymptotic optimality in mean does not imply weak asymptotic optimality.
16. [C32] Derive an ε-optimal equivalent of the notions of optimality discussed in this chapter. How do they differ from the non ε-optimal versions? When are they equivalent?

The bracketed numbers such as [C20] are the book's difficulty ratings for each exercise (roughly, "C" for a problem requiring creative/compositional reasoning, with the number indicating relative difficulty/length).

8.5 History and References I

Notions of optimality. This chapter is based on material from [LH15b] and [Lei16b, Chp. 5]. Universal Bayesian agents were introduced in [Hut02c], which showed that $\mathrm{AI}\xi$ is **Self-Optimizing** and **Pareto-Optimal**. A comprehensive review of many of the notions of optimality discussed in this chapter and beyond can be found in [Hut05b, LH15b, Lei16b].

Related to regret and asymptotic optimality are the **PAC** and **sample-complexity** based optimality notions. Near-optimal PAC bounds for finite discounted MDPs based on Upper Confidence Reinforcement Learning (**UCRL**) [JOA10] have been proven in [LH12, LH14c]. The first PAC result in *general* reinforcement learning was given in [LHS13b], with additional work being done in the case of feature abstractions being utilized in [RLDG22]. A simple but very general Bayesian regret bound covering and unifying many special cases was presented in [LVRD$^+$21].

8.5 History and References II

Asymptotic optimality. The difficulty of asymptotic optimality was first demonstrated in [Ors10], where it was shown that AIXI is *not* asymptotically optimal. These results were extended in [LH11a], where the authors defined two notions of asymptotic optimality and showed that under certain conditions these can(not) be satisfied. One of the difficulties with asymptotic optimality is that to achieve it, an agent needs to **explore** enough to gain sufficient information [RVR14, RCV16].

There are downsides to this exploration: in particular, it was shown that if an agent explores *enough* to be asymptotically optimal, then it also explores enough to **die with certainty** [CHC21]. Algorithms able to achieve both asymptotic-optimality and regret-based optimality results, as well as the connection between the two, are presented in [KLVS21, LLOH16, Lei16b]. The existence of a policy able to achieve **strong** asymptotic optimality was proven in [CCH19].

Summary

- ν-**optimality / ε-optimality**: the ideal but unrealistic gold standard – optimal in one (possibly unknown) true environment.
- **Asymptotic optimality** (strong $\Rightarrow$ in mean, in probability $\Rightarrow$ weak): eventually optimal, in various probabilistic senses; connected to **sublinear regret** via Theorem 8.1.13 under recoverability.
- **Bayesian optimality**: optimal w.r.t. a mixture over an environment class; AIXI is Bayes-optimal in $\mathcal{M}_{sol}$ with the universal prior; the "best of both worlds" criterion.
- **Regret minimization**: a finite-horizon, short-term notion, in tension with asymptotic optimality (Example 8.1.10).
- **Pareto optimality**: too weak alone (vacuous for $\mathcal{M} \supseteq \mathcal{M}_{comp}$, Theorem 8.1.19), but its **balanced** refinement coincides exactly with Bayes-optimality.
- **Bad priors** (dogmatic, indifference) show that Bayes-optimality's behaviour *depends heavily* on the prior – motivating the search for *natural* universal priors.
- **Impossibility results** (Theorems 8.3.1, 8.3.2) show determinism and full generality are each obstacles to asymptotic optimality – but stochastic exploration can restore it (Chapter 9).